

# Hadi Lashkari Ghouchani

## Software Engineer

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### Summary

Software engineer with 21+ years of development experience, including 11 years of professional Android development and, most recently, mobile CI/CD and developer-experience engineering. Skilled in software architecture, algorithm design, and applied machine learning.

- Builds scalable, maintainable, and efficient systems, from low-level Android internals to app architecture
- Automates mobile delivery with trunk-based release pipelines, reproducible Nix builds, and AI-agent workflows
- Advocates SOLID principles, single source of truth, pure functions, and stateless design
- Applies machine learning to problems that make a real difference in people's lives
- Taught algorithms and programming at university level
- Continuous learner and open-source contributor
- Makes technical decisions proactively and transparently

Looking to join a creative, motivated team working on challenging technical problems.

### Experience

#### Senior Mobile Developer

at Bell Canada April, 2026 - Present Toronto, Canada

Bell Canada is a Canadian telecommunications company

Mission:

- Took ownership of GitLab CI/CD and reversed a declining trend in pipeline job success rate, raising it from 18% to 31% (a 72% relative improvement) within six months
- Designed and implemented trunk-based development pipelines for Android and iOS across multiple repositories, automating release-branch creation, cherry-picks, publishing, and release-candidate promotion
- Built a multi-agent CI workflow that turns Jira tickets into merge requests, with planner, coder, and reviewer agents committing an initial implementation
- Automated DTO and mapper generation from APIDog API specs in CI, eliminating hand-written boilerplate from developer merge requests
- Restructured merge request pipelines into fast checks, required heavy jobs, and optional jobs, and strengthened automated code verification, shortening feedback loops and raising confidence before merge
- Made builds and emulator runs reproducible by wrapping them in Nix flakes
- Migrated GitLab jobs to custom, task-specific Linux containers, improving performance, isolation, and reproducibility
- Reduced GitLab SAST security findings to zero across multiple projects
- Pinned dependencies, tuned CI/CD caching, and established architecture and release-readiness practices
- Maintain and tune the EC2 instances running CI/CD for reliability and performance

Technologies:

- Android, Kotlin Multiplatform, Gradle Plugin
- iOS, Swift, XCode, XCodebuild
- CI/CD, Gitlab's Configurations, Gitlab SAST, Podman, Nix, AWS EC2

#### Senior Mobile Developer

at Block Inc. - Bitkey project(via Touchlab) November, 2025 - Feb, 2026 Toronto, Canada

Block, Inc. is an American technology company and a financial services provider for consumers and merchants. The Bitkey is a project in Block Inc. that I directly work on. Bitkey is a digital wallet for bitcoin, available in the U.S. and 95 other countries.

Mission:

- Worked on multi backup project
- Fixed flaky tests, including unit and functional tests by solving multiple race conditions and state management problems
- Worked on the new onboarding flow

Technologies:

- Kotlin Multiplatform/KMP, Compose Multiplatform/CMP, Gradle Plugin
- iOS, Swift, XCode
- Rust binding, UniFFI
- Ktor, Anvil, SqlDelight, Kotest
- Github, Linear, Notion, BuildKite, etc.

#### Lead Android Developer

at CIBC (via Vision Talent) March, 2025 - November 2025 Toronto, Canada

The Canadian Imperial Bank of Commerce is a Canadian multinational banking and financial services corporation headquartered at CIBC Square in the Financial District of Toronto, Ontario.

Mission:

- Rewriting features in the newly developing CIBC mobile apps
- Improved the shared CI pipeline build-time by almost 50%, mostly by using build caches
- Initiated and Implemented memory management by applying structural DI scopes to release resources, ViewModels(in MVVM architecture), and their state
- Working on remote FeatureFlag functionality
- Migrated a Confluence's scraping script to use Selenium

Technologies:

- Kotlin Multiplatform/KMP, Android, Compose Multiplatform/CMP, Gradle Plugin, code generator
- iOS, Swift, XCode, SwiftUI
- Ktor, Koin, Compose Navigation, etc.
- Atlassian Jira, Atlassian Confluence, Github, Jenkins, etc.

#### Software Engineer -> Productivity Engineer

at Bell Canada (via MobileLIVE) Aug, 2022 - Nov, 2024 Toronto, Canada

Promoted to Productivity Engineer.

Mission:

- Resolved VPN network proxy and self-signed certificate issues by developing an automated installation script and Compose Multiplatform app, resulting in a 164% increase in build count on the Gradle Enterprise(Develocity) dashboard.
- Initiated and implemented Gradle Conventional Plugins to reduce code duplication and errors. Established Single Source of Truth (SSOT) for external library versions, significantly reducing runtime crashes caused by version conflicts.
- Spearheaded the adoption of Merge Trains and Merge Request Pipelines in GitLab, creating a Proof of Concept (POC) workflow that demonstrably improved code quality and streamlined the integration process.
- Optimized test execution, reducing incremental runtime by 52% and improving overall development efficiency.
- Served as the primary resource for build-related training and support across a team of 50+ Android developers, enhancing team productivity and knowledge sharing.
- Implemented a custom No-Framework dependency injection solution across multiple app and library repositories, enhancing modularity, testability, and maintainability while minimizing build times by avoiding code generation.
- Integrated Detekt to verify code quality in the merge request's pipelines.

Additional Technologies:

- Develocity(Gradle Enterprise), Gradle Conventional Plugin, Detekt
- Kotlin Multiplatform, Compose Multiplatform

#### Software Engineer

at Bell Canada (via MobileLIVE) Apr. 2021 - Aug. 2022 Toronto, Canada

MobileLIVE: A Canadian tech-service provider specializing in digital transformation solutions. Bell Canada: Canada's largest telecommunications company, offering mobile services (Bell Mobility, Virgin Mobile), television (Bell Satellite TV, Bell Fibe TV), and internet services across the country.

Mission:

- Implemented a Request for Comments (RFC) process for technical decision-making, streamlining the evaluation and adoption of new ideas across the organization.
- Led regular knowledge-sharing discussions across the team on SOLID principles, modularization best practices, feature flag implementation, and monorepo architecture, elevating the team's overall code quality and architectural design skills.
- Conceived and executed an automated system for cloning and building source code across a multi-repo setup, significantly reducing version conflicts and enhancing code visibility for efficient debugging.

Technologies:

- Kotlin Coroutines / Mockk, Mockito, JUnit, etc.
- Agile, Sprint, Lane based branching strategy, CI/CD
- Atlassian Jira, Atlassian Confluence, Gitlab, Jenkins, Slack, Branch.io, inVision, Dynatrace, etc.

#### Staff Android Developer

at iFood (via Letgo) Oct. 2020 - Mar. 2021 Barcelona, Spain

iFood is a leading food delivery platform, dominating markets in Brazil, Colombia, and Mexico. It's the market leader in Latin America with 50+ million Android app installations. Mission-critical achievement: Initiated and executed a strategy to refactor legacy code out of the root module, resulting in a 55% decrease in incremental build time and improved overall system architecture.

#### Senior Android Developer

at Letgo: Leading Classified Marketplace Oct. 2019 - Mar. 2021 Barcelona, Spain

One of the largest classified marketplaces in the United States at the time. Market leader in Turkey with 15+ million Android app installations. Mission-critical achievement: Optimized feed loading performance, reducing launch time by over 60 seconds.

#### Lead Android Developer

at Sheypoor: Top-Tier Classified Marketplace Jan. 2018 - Jan. 2019 Tehran, Iran

Impressive user base: 13+ million overall users, 1.5+ million daily active users. High engagement: 344+ million sessions. Key Achievement: Spearheaded initiatives that tripled daily active users from 500,000 to 1.5+ million within one year.

#### Senior Android Developer

at Myket: Top-Tier App Store and Social Network Sep. 2015 - Des. 2017 Tehran, Iran

#### Java and C# developer

at Soha System Mar. 2015 - Aug. 2015 Tehran, Iran

### Public Contributions

- 🔗 2026 Tabular center: State machines whose declaration is the transition matrix
- 🔗 2026 Type Neural Network: A machine learning architecture based on the Type Mechanics
- 🔗 2026 MFFT benchmark: A benchmark among matmul implementations with focus on MFFT
- 📄 2025 Post: Type Mechanics
- 📄 2025 Post: Geometrical Probability
- 📄 2024 Post: Matrix Fast Fourier transform(MFFT) - An optimization for Matrix Multiplication
- 📄 2024 C Numbers: This is a collection of number types, including floating numbers and p-adic numbers.
- 📄 2024 Post: Not All codes are liabilities
- 🔗 2024 City of Toronto RAG GPT: The project received the second position award in the TMLS Hackathon.
- 🔗 2024 dragon detector: A simple ML model to detect dragon in images
- 🔗 2024 Nix Effect pod: Modularize effects of all kinds, and stay in-control, with NixOS and podman.
- 🔗 2023 Contributing to nixpkgs/androidenv: Nix is a language that is designed to have reproducible output.

### Education

|                                 |      |            |             |                 |
|---------------------------------|------|------------|-------------|-----------------|
| Sharif University Of Technology | M.Sc | in Physics | 📅 Feb. 2010 | 📍 Tehran, Iran  |
| Payame Noor University          | B.Sc | in Physics | 📅 Jul. 2007 | 📍 Mashhad, Iran |

### Tools / Skills

#### Kotlin Multiplatform/KMP

Compose Multiplatform/Android/iOS/Desktop  
Advocate for independent UI/UX from platform  
Main Contributor of Log4K  
Main Contributor of LifecycleHandler

#### Android

MVVM/Android Architecture Components/JetPack  
Anvil/Dagger 2/RxJava 2/Retrofit 2/...  
Kotlin Multiplatform/Kotlin Coroutines  
custom Kotlin Compiler Plugins/...  
custom KSP plugins/...  
Jetpack Compose/View Binding/...  
Contributor of NixOS's Android Env

#### iOS

Swift/SwiftUI

#### Software Architecture

Clean Architecture/Hexagonal Architecture  
Unidirectional data flow Architecture  
Stateless Architecture  
Logic Tree Architecture  
MVVM/MVI/MVP/RMVVM

#### Principle

Clean Code/SOLID  
Object Oriented Programming  
Functional Programming  
Isolation/Reproducibility  
Stable Dependencies Principle  
Stable Abstractions Principle

#### Design Pattern

Factory/Builder/Singleton/...  
Adapter/Facade/Proxy/Visitor/Delegate/...  
Strategy/Chain of responsibility/...

#### Modularization Strategies

Applying DIP/Decrease incremental build time  
Code isolation to reduce dependencies

#### Continuous Integration/Deployment

Jenkins, GitLab CI, Fastlane

#### Android Performance Optimization

memory management/threading/profiling tools

#### Android Security

encryption/secure networking/permissions model

#### Android Test and Verification

Unit/Integration/Mockito/Espresso/JaCoCo/...  
Detekt plugins/Konsist

#### Android Build System

ProGuard/R8/custom Gradle plugins

#### Build Tool

Nix/Gradle/Cargo/Sbt/Maven/Ant/Make/...

#### Command Line Tool

tmux/bash/zsh/git/grep/sed/netstat/awk/podman/...  
adb/sdkmanager/avdmanager/xcodebuild/...

#### Rust

actix/rusqlite/tokio/thiserror/crossbeam  
tonic/prost/openssl/flatbuffers/thrift  
Rc/Arc/Mutex  
FFI

#### Podman/Docker

Buildkit/Compose/Cache

#### Nix

NixOS/Nix Darwin/Home Manager  
mkDrivation/mkShell

#### Programing Language

Kotlin/Rust/Python/Java/Scala  
Groovy/Dart/JavaScript/Ruby...

#### Workflow

Trunk Based development/Github flow/Git Flow  
Cross-functional team

#### Operating System

Linux(NixOS/Fedora/Ubuntu/Debian), MacOS

### Achievements and Recognition

- 2026 Had an offer from Bell Canada
- 2025 Had an offer from Block Inc. via Touchlab
- 2025 Had an offer from CIBC via Vision Talent
- 2024 Had an offer from Accenture via Experis
- 2024 First position in National Health Datathon
- 2024 Second position in AWS Summit's Gameday
- 2024 Second position in Toronto Machine Learning #TMLS Hackathon
- 2020 Had an offer from MobileLIVE, Toronto, Canada
- 2020 Had an offer from iFood, Remote
- 2019 Had an offer from Kik, Kitechener, Canada
- 2019 Had an offer from Letgo, Barcelona, Spain
- 2019 Had an offer from CRITICAL Techworks, BMW, Porto, Portugal
- 2019 Had an offer from The Next Big Thing, Berlin, Germany
- 2018 Received result-oriented badge in Sheypoor
- 2007 Ranked 44th among more than five thousand participants in the National Graduate Entrance Exam in Physics

### Language

English: Advance  
Farsi: Native

The latest version of this resume is available in [github](#)  
Built on September 28, 2026